

# Malware Analysis Report For pe-Windows-x86-cmd

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## Binary Type

PE (Portable Executable File) (Approved)

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## Analyzing File

- **File Name:** pe-Windows-x86-cmd
  - **File Extension:**
  - **File Type:** EXE (Executable File).
  - **MD5 Hash:** e52110456ec302786585656f220405eb
  - **SHA256 Hash:** 785c974152976525e46b032c30378e457d069f30ac3f0fe9613e5e142ce7e8b2
- 

## File Identification

File detected as a valid Windows executable (DLL or EXE).

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## PE File Analysis

PE File Analysis: True

### PE File Imports

- exit
- \_getch
- iswdigit
- wcstol
- \_vsnwprintf
- \_controlfp
- ?terminate@@@YAXXZ
- except\_handler4\_common
- \_setapp\_type
- pfmode
- pcommode
- \_\_setusermatherr
- \_amsgexit
- \_initterm
- \_XcptFilter
- \_exit
- \_cexit
- \_\_getmainargs
- calloc
- free
- \_wcslwr
- qsort
- \_dup2
- \_dup

- *openosfhandle*
- *\_close*
- *swscanf*
- *\_ultoa*
- *\_pipe*
- *\_setmode*
- *wcsncmp*
- *iswxdigit*
- *\_wtol*
- *time*
- *srand*
- *fflush*
- *getosfhandle*
- *\_setjmp3*
- *wcsstr*
- */local/unwind4*
- *\_errno*
- *wcstoul*
- *iswalpha*
- *rand*
- *wcsrchr*
- *memcpy*
- *printf*
- *\_iob*
- *fprintf*
- *towlower*
- *realloc*
- *setlocale*
- *\_wcsupr*
- *\_wopen*
- *ferror*
- *fgets*
- *feof*
- *\_pclose*
- *memmove*
- *wcschr*
- *\_tell*
- *iswspace*
- *memset*
- *wcsspn*
- *towupper*
- *longjmp*
- *\_wcsnicmp*
- *\_wcsicmp*
- *RtlNtStatusToDosError*
- *NtSetInformationProcess*
- *NtQueryInformationProcess*
- *RtlFreeHeap*
- *NtFsControlFile*
- *RtlDosPathNameToNtPathName\_U*
- *NtQueryInformationToken*
- *NtClose*

- NtOpenProcessToken
- NtOpenThreadToken
- GetDateFormatW
- UnhandledExceptionFilter
- GetCurrentProcess
- GetSystemTimeAsFileTime
- GetCurrentProcessId
- GetTickCount
- QueryPerformanceCounter
- GetModuleHandleA
- SetUnhandledExceptionFilter
- Sleep
- InterlockedExchange
- LoadLibraryExA
- InterlockedCompareExchange
- FreeLibrary
- DelayLoadFailureHook
- CreateHardLinkW
- CreateSymbolicLinkW
- GetVolumePathNameW
- GetThreadLocale
- GetSystemInfo
- SetProcessAffinityMask
- ResumeThread
- FindFirstFileExW
- DeviceIoControl
- FindFirstStreamW
- FindNextStreamW
- GetDiskFreeSpaceExW
- CompareFileTime
- RemoveDirectoryW
- GetCurrentDirectoryW
- SetCurrentDirectoryW
- TerminateProcess
- WaitForSingleObject
- GetExitCodeProcess
- CopyFileW
- SetFileAttributesW
- SetEndOfFile
- DeleteFileW
- SetFileTime
- CreateDirectoryW
- FillConsoleOutputAttribute
- SetConsoleTextAttribute
- ScrollConsoleScreenBufferW
- FormatMessageW
- GetACP
- DuplicateHandle
- FlushFileBuffers
- HeapReAlloc
- HeapSize
- VirtualAlloc

- VirtualFree
- GetCurrentThreadId
- OpenThread
- HeapSetInformation
- GetFileAttributesExW
- GetDriveTypeW
- InitializeCriticalSection
- SetConsoleCtrlHandler
- GetWindowsDirectoryW
- GetVersion
- GetModuleFileNameW
- ExpandEnvironmentStringsW
- CancelSynchronousIo
- EnterCriticalSection
- LeaveCriticalSection
- GetVolumeInformationW
- SearchPathW
- WriteFile
- SetFilePointerEx
- GlobalAlloc
- GlobalFree
- MoveFileW
- SetConsoleTitleW
- LocalFree
- MoveFileExW
- GetConsoleTitleW
- GetFileAttributesW
- NeedCurrentDirectoryForExePathW
- GetBinaryTypeW
- SetFilePointer
- lstrcpwW
- lstrcpwW
- HeapFree
- GetProcessHeap
- SetThreadLocale
- GetProcAddress
- GetModuleHandleW
- VirtualQuery
- HeapAlloc
- CloseHandle
- MultiByteToWideChar
- ReadFile
- WriteConsoleW
- FillConsoleOutputCharacterW
- SetConsoleCursorPosition
- ReadConsoleW
- GetConsoleScreenBufferInfo
- GetStdHandle
- GetFileType
- GetLastError
- WideCharToMultiByte
- GetFileSize

- FlushConsoleInputBuffer
- GetCPInfo
- GetConsoleOutputCP
- CmdBatNotification
- CreateFileW
- FindClose
- FindNextFileW
- FindFirstFileW
- GetFullPathNameW
- GetUserDefaultLCID
- SetLocalTime
- SystemTimeToFileTime
- GetSystemTime
- FileTimeToSystemTime
- GetLocaleInfoW
- FileTimeToLocalFileTime
- GetTimeFormatW
- GetLocalTime
- FreeEnvironmentStringsW
- GetEnvironmentStringsW
- SetEnvironmentVariableW
- SetEnvironmentStringsW
- SetConsoleMode
- GetConsoleMode
- GetCommandLineW
- GetEnvironmentVariableW
- SetErrorMode
- GetVDMCurrentDirectories
- RegCloseKey
- SetLastError
- RegQueryValueExW
- RegOpenKeyExW
- RegDeleteValueW
- RegDeleteKeyExW
- ReadProcessMemory
- LoadLibraryW
- QueryFullProcessImageNameW
- GetConsoleWindow
- CreateProcessW
- GetStartupInfoW
- DeleteProcThreadAttributeList
- UpdateProcThreadAttribute
- InitializeProcThreadAttributeList
- RegSetValueExW
- RegCreateKeyExW
- BrandingFormatString

## PE File Exports

No Exported functions were found.

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# String Analysis

## ASCII Strings

- !This program cannot be run in DOS mode.
- `.data
- @.reloc
- msvcrt.dll
- ntdll.dll
- KERNEL32.dll
- api-ms-win-core-processthreads-l1-1-0.DLL
- WINBRAND.dll
- QQSVWj
- ,j%\_f;
- J3t/95tA
- Jj/hl\$
- Jj2hh\$
- Jj1hT'
- Jj0hx'
- ?"u3GG
- Jj Zf;
- JYY\_^[
- SetConsoleInputExeNameW
- IsDebuggerPresent
- CopyFileExW
- Jj [Sh
- SetThreadUILanguage
- 9:tq9]
- u.WhpF
- F<@@Ph
- FFf9>u
- :tXAAC
- E\$WhT1
- SSSSSWj
- @
- QRPh<#
- SVWj=Xf
- 9x t99=
- Juz@PW
- JWj"\_f;
- SSSSVh
- J )#j@
- JPWPVhl
- APerformUnaryOperation: '%c'
- APerformArithmeticOperation: '%c'
- ItGlt9
- LGj:GXf
- YFj:FXf
- FFj\Xf
- Nj0NXf
- Null environment
- (j:Xj.f

- Jt\j/P
- u"Sh7#
- Jt\j/P
- u"Sh8#
- tbf97tj
- NtQueryInformationProcess
- PSWSSS
- AjcAZf
- Aj)AZf
- HuhSVhx'
- jSVhT'
- ;u7SVh
- SVWt#j
- tAHt3HHt\$
- SSSSSS
- ?"u0j"Yf
- FFGGCC
- tZHt9Ht
- QRPh#
- ADVAPI32.dll
- USER32.dll
- SHELL32.dll
- MPR.dll
- RevertToSelf
- SaferRecordEventLogEntry
- ImpersonateLoggedOnUser
- SaferCloseLevel
- SaferComputeTokenFromLevel
- SaferIdentifyLevel
- RegEnumKeyW
- RegSetValueW
- CreateProcessAsUserW
- LookupAccountSidW
- GetSecurityDescriptorOwner
- GetFileSecurityW
- MessageBeep
- ShellExecuteExW
- SHChangeNotify
- WNetCancelConnection2W
- WNetGetConnectionW
- WNetAddConnection2W
- WINBRAND.dll
- KERNEL32.dll
- ntdll.dll
- msvcrt.dll
- \_getch
- iswdigit
- wcstol
- \_vsnwprintf
- \_controlfp
- ?terminate@@YAXXZ
- except\_handler4\_common

- `_setapp_type`
- `pfmode`
- `pcommode`
- `__setusermatherr`
- `amsgexit`
- `_initterm`
- `_XcptFilter`
- `_cexit`
- `__getmainargs`
- `calloc`
- `_wcslwr`
- `openosfhandle`
- `_close`
- `swscanf`
- `_ultoa`
- `_setmode`
- `wcsncmp`
- `iswxdigit`
- `fflush`
- `getosfhandle`
- `_setjmp3`
- `wcsstr`
- `/local/unwind4`
- `_errno`
- `wcstoul`
- `iswalpha`
- `wcsrchr`
- `memcpy`
- `printf`
- `fprintf`
- `tolower`
- `realloc`
- `setlocale`
- `_wcsupr`
- `_wpopen`
- `ferror`
- `_pclose`
- `memmove`
- `wcschr`
- `iswspace`
- `memset`
- `wcsspn`
- `towupper`
- `longjmp`
- `_wcsnicmp`
- `_wcsicmp`
- `RtlNtStatusToDosError`
- `NtSetInformationProcess`
- `NtQueryInformationProcess`
- `RtlFreeHeap`
- `NtFsControlFile`
- `RtlDosPathNameToNtPathName_U`



- NtQueryInformationToken
- NtClose
- NtOpenProcessToken
- NtOpenThreadToken
- GetDateFormatW
- UnhandledExceptionFilter
- GetCurrentProcess
- GetSystemTimeAsFileTime
- GetCurrentProcessId
- GetTickCount
- QueryPerformanceCounter
- GetModuleHandleA
- SetUnhandledExceptionFilter
- InterlockedExchange
- LoadLibraryExA
- InterlockedCompareExchange
- FreeLibrary
- DelayLoadFailureHook
- CreateHardLinkW
- CreateSymbolicLinkW
- GetVolumePathNameW
- GetThreadLocale
- GetSystemInfo
- SetProcessAffinityMask
- ResumeThread
- FindFirstFileExW
- DeviceIoControl
- FindFirstStreamW
- FindNextStreamW
- GetDiskFreeSpaceExW
- CompareFileTime
- RemoveDirectoryW
- GetCurrentDirectoryW
- SetCurrentDirectoryW
- TerminateProcess
- WaitForSingleObject
- GetExitCodeProcess
- CopyFileW
- SetFileAttributesW
- SetEndOfFile
- DeleteFileW
- SetFileTime
- CreateDirectoryW
- FillConsoleOutputAttribute
- SetConsoleTextAttribute
- ScrollConsoleScreenBufferW
- FormatMessageW
- GetACP
- DuplicateHandle
- FlushFileBuffers
- HeapReAlloc
- HeapSize

- VirtualAlloc
- VirtualFree
- GetCurrentThreadId
- OpenThread
- HeapSetInformation
- GetFileAttributesExW
- GetDriveTypeW
- InitializeCriticalSection
- SetConsoleCtrlHandler
- GetWindowsDirectoryW
- GetVersion
- GetModuleFileNameW
- ExpandEnvironmentStringsW
- CancelSynchronousIo
- EnterCriticalSection
- LeaveCriticalSection
- GetVolumeInformationW
- SearchPathW
- WriteFile
- SetFilePointerEx
- GlobalAlloc
- GlobalFree
- MoveFileW
- SetConsoleTitleW
- LocalFree
- MoveFileExW
- GetConsoleTitleW
- GetFileAttributesW
- NeedCurrentDirectoryForExePathW
- GetBinaryTypeW
- SetFilePointer
- lstrcpw
- lstrcpw
- HeapFree
- GetProcessHeap
- SetThreadLocale
- GetProcAddress
- GetModuleHandleW
- VirtualQuery
- HeapAlloc
- CloseHandle
- MultiByteToWideChar
- ReadFile
- WriteConsoleW
- FillConsoleOutputCharacterW
- SetConsoleCursorPosition
- ReadConsoleW
- GetConsoleScreenBufferInfo
- GetStdHandle
- GetFileType
- GetLastError
- WideCharToMultiByte

- GetFileSize
- FlushConsoleInputBuffer
- GetCPInfo
- GetConsoleOutputCP
- CmdBatNotification
- CreateFileW
- FindClose
- FindNextFileW
- FindFirstFileW
- GetFullPathNameW
- GetUserDefaultLCID
- SetLocalTime
- SystemTimeToFileTime
- GetSystemTime
- FileTimeToSystemTime
- GetLocaleInfoW
- FileTimeToLocalFileTime
- GetTimeFormatW
- GetLocalTime
- FreeEnvironmentStringsW
- GetEnvironmentStringsW
- SetEnvironmentVariableW
- SetEnvironmentStringsW
- SetConsoleMode
- GetConsoleMode
- GetCommandLineW
- GetEnvironmentVariableW
- SetErrorMode
- GetVDMCurrentDirectories
- RegCloseKey
- SetLastError
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- RegOpenKeyExW
- RegDeleteValueW
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- GetConsoleWindow
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- GetStartupInfoW
- DeleteProcThreadAttributeList
- UpdateProcThreadAttribute
- InitializeProcThreadAttributeList
- RegSetValueExW
- RegCreateKeyExW
- BrandingFormatString
- cmd.pdb
- CMD Internal Error %s
- 
- 
-

- [illegible]

- :1:7:=:C:J:]e:m:r:w:|;
- <;
- =2=\_d={=
- ='>=>K>W>\_>~>
- 7\*7=7B7I7`7e7I7
- 94:A:Y:
- ;,,Z;b;|;
- <.
- ?0Y0m0
- 8-8Q8k8
- 839c9v9L:j:x:
- :';1;\;';~;
- =/=6=C=Y='=
- 0!0)0B0I0V0v0
- 1&1=1S1X1
- 4)595z5
- 798Q8g9p9}9
- 9>:M:y:
- ;9;?;R;
- 
- ?I?V?\?I?}?
- 0#03090y0
- 141C1g1r1z1
- 2@2E2K2Q2f2t2
- 3"3/363K3
- 5K5T5f5
- 5'6B6s6
- 7K7e7I7
- :6:C:H:N:T:g:n:s:
- :Y;l;s;
- <#<'<=
- =)=B=Z=b=
- >&>:>J>S>
- ? ??:?a?m?u?}?
- 1%1/14191M1V1[1h1y1
- 2\*2B2O2U2^2}2
- 3\$3\*32383E3m3
- 3\$4(4.454\4i4v4
- 6\*727Y7a7|7
- 888N8X8
- 9%9d9t9z9
- ;-;V;e;
- 
- 0"0X0w0
- 1&151=1V1o1z1
- 6,6C6\_6
- 677@7I7U7f7y7
- 8 8;8h8
- 9 9(9t9
- ;R;Y;';
- <+<4
- = =%=D=

- >&>6>~>
- >Q?Y?p?}??
- 021H1c1q1
- 2A2]2e2
- 3.4;4@4R4`4r4
- 4,5?5F5
- 7-8J8a8
- :7<`
- =A>[>n>
- ?#?.?3?B?
- 13191A1H1R1
- 2:3@3W3]3f3
- 7,808c8
- 9=9B9T9g9v9
- 9&:E:S:X:t:
- :\$;D;];
- 
- ?0?D?N?\?
- 5\*6=6D6
- ?5?B?Q?V?
- ;0Z0=1X1
- 7&7@7X7
- 7\$8(8:8d8s8
- 9,9<9I9W9
- 161D1V1a1o1
- 2I2p2z2
- 3/4<4F4a4
- 5C5I5s5
- 7L8\8d8j8
- 9&9/959;9X9z9
- 9 :\_i:
- <'=F=g=
- 3\_4x4A5Y5
- ;=
- =&=:N=y=
- 1>2\2u2
- 2\$3K3u3
- 4;4Y4\_4g4y4
- 4\$52595I5U5h5
- 686K6f6
- );U;\_;
- &?B?
- 0%1+1P1
- ;!<-
- 4+4O4\4m4
- 595?5`5
- 7\*717?7F7S7
- 8)9<9G9p9
- :!;0;8;B;O;
- <)
- <'===C=X=x=
- 525?5L5I5

- 5#6+6l6u6z6
- >%>0>?>P>
- ?\$?.?8?
- 000:0n0
- 2U3\3y3
- 3&4-4<5\5z5
- 5'6;6A6j6
- 9V9j9j9
- 3(3C3l3j3j3
- 4"4'4,484
- 415;5Q5H6T6j6g6
- >1>]>p>y>
- ?,?Q?X?\_?f?m?}??
- 0)050K0\0`0d0h0l0p0t0x0|0
- 0:1C1o1u1{1
- 3(3/3i3
- 4%4,4Z4
- 5,5>5M5a5n5z5
- 7!7'7:7h7q7}7
- 8\$888d8B9f9
- ;\$<.
- ??:?B?H?d?r?
- 0'050;0s0
- 7!747A7P7n7
- 7/8Q8b8m8
- :\$.?:z:
- <==M=[=
- 5,5g5t5
- 657<7L7R7g7o7v7|7
- 7 8=8L8h8v8
- 9O9\9x9
- :A:O:v:
- =\$?2?6?=?C?O?]?f?y?
- 0'0K0l0q0
- 1&1U1o1
- 262U2|2
- 5'585E5R5\_5j5
- 7\*787F7T7b7
- 9 ;(i;q;
- =0=9=f=v=
- >/>>>G>O>
- 0"050j0z0
- 1l2O2^2c2k2
- 3\*353:3E3
- 4b4i4p4
- 8H8W8y8
- 0'04090S0u0{0
- 273f3n3y3
- 8/969@9O9[9o9u9j9
- =0=T=e=q=
- >%?D?Q?k?
- 2#323V3u3

- 7h9m9~9
- ;E;K;};
- 
- <<=h=x=
- 3&3P3)4
- 6%6>6J6U6d6
- 7!7,7G7`7g7
- 8]9d9k9w9
- 92:8:c:
- ;.<9<<
- 0\$0V0w0
- 262F2s3
- 516=6s6
- 7=7R7p7u7}7
- 929J9T9
- 9W:a:f:u:
- ?9?N?\_?
- 141;1A1F1O1V1[1c1i1
- 0/191J1[1
- 3.373?3G3N3S3]3e3p3z3
- 4#4/4x4
- 7\$787E7M7[7a7g7
- ;T;Z;d;o;
- =7=@=L=U=h=t=
- =f>u>|>
- ?%?+?e?
- 0#01070@0F0
- 0.1=1E1Q1t1~1
- 1<2D2^2
- 223:3l3Q3Y3
- 6:7G7R7
- :(k:y:
- ;o;y;~;
- 0-0K0Y0j0~0
- 081a122
- 565C5Q5n5u5
- 
- =\$/=:=E=
- \$090@0
- 0+121l1T1Z1
- 252<2D2l2^2c2j2u2{2
- 8"8H8l8
- :9:H:O:
- 1&1,12181>1D1J1P1V1\1b1h1o1w1
- 2(2-272A2K2U2\_2i2s2}2
- 0 0\$0(0,040<0@0H0L0P0
- @8D8X8\8p8t8
- 90949H9L9`9d9x9|9
- :.\$:8:<:P:T:h:l:
- ;(,;,@;D;X;\;p;t;
- 
- = =\$=8=<=P=T=h|=



## UTF-8 Strings

No detected.

## UTF-16LE Strings

- CMD.EXE
- ()|&=,;"
- CMD CMDLINE
- RANDOM
- COPYCMD
- \XCOPY.EXE
- KERNEL32.DLL
- AutoRun
- PathCompletionChar
- CompletionChar
- DefaultColor
- DelayedExpansion
- EnableExtensions
- DisableUNCCheck
- Software\Microsoft\Command Processor
- MM/dd/yy
- DisableCMD
- Software\Policies\Microsoft\Windows\System
- 0123456789
- SCRIPT
- ENABLEEXTENSIONS
- ENABLEDELAYEDEXPANSION
- DISABLEEXTENSIONS
- cmd.exe
- ??????????.???
- BELOWNORMAL
- ABOVENORMAL
- AFFINITY
- Unknown
- /K %s
- /D /c"
- %d.%d.%04d
- DIRCMD
- tokens=
- delims=
- useback
- usebackq
- SHARED
- SEPARATE
- REALTIME
- NORMAL
- DISABLEDELAYEDEXPANSION
- pushd
- mkdir
- rmdir
- chdir

- dd/MM/yy
- yy/MM/dd
- HH:mm:ss t
- %WINDOWS\_COPYRIGHT%
- Ungetting: '%s'
- REM /?
- FOR /?
- GetToken: (%x) '%s'
- [...]
- Software\Classes
- System
- Application
- NTDLL.DLL
- %x %c
- Redir:
- \* Unknown type: %x
- Args: '%s'
- Cmd: %s Type: %x
- %s (%s) %s
- 
- DEFINED
- PROMPT
- RENAME
- VERIFY
- MKLINK
- COMSPEC
- PATHEXT
- PROMPT
- %04X-%04X
- ENDLOCAL
- SETLOCAL
- \CMD.EXE
- =ExitCode
- fdpxsatz
- ERRORLEVEL
- =,;+/\]
- CMDEXTVERSION
- <>+-%/%)|^&=,
- =ExitCodeAscii
- %2d%s%02d%s%02d%s%02d
- %02d%s%02d%s
- %02d%s%02d%s%02d
- %s %s%s
- (%s) %s
- %s %s
- .COM;.EXE;.BAT;.CMD;.VBS;.JS;.WS;.MSC
- \Shell\Open\Command
- &(){}^=;!%'+'~
- IDI\_APPICON
- VS\_VERSIONINFO
- StringFileInfo
- 040904B0

- CompanyName
- Microsoft Corporation
- FileDescription
- Windows Command Processor
- FileVersion
- 6.1.7600.16385 (win7\_rtm.090713-1255)
- InternalName
- LegalCopyright
- Microsoft Corporation. All rights reserved.
- OriginalFilename
- Cmd.Exe
- ProductName
- Microsoft
- Windows
- Operating System
- ProductVersion
- 6.1.7600.16385
- VarFileInfo
- Translation

Extracted URLs

- <http://schemas.microsoft.com/SMI/2005/WindowsSettings>

Base64 Strings

No valid Base64-encoded strings found.

File Entropy Analysis

- **Entropy:** 4.59
- **Status:** This file likely contains Standard Text or human-readable text.

VirusTotal Analysis

Scan Date: 2024-01-12 19:36:58

Scan Results

- Malicious: 1
- Suspicious: 0
- Undetected: 68
- Harmless: 0
- Timeout: 0
- Confirmed-timeout: 0
- Failure: 0
- Type-unsupported: 4

Community Votes

- Harmless: 0
- Malicious: 0

Detailed Engine Results

| Engine Name | Detection Category | Detection           |
|-------------|--------------------|---------------------|
| Bkav        | malicious          | W32.AIDetectMalware |

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